STUDENT ATTENDANCE MANAGEMENT SYSTEM

Project Report

Technology: Java Swing, JDBC and MySQL

Submitted by: _____

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1. Abstract

The Student Attendance Management System is a desktop application designed to digitize recording and monitoring student attendance. Manual registers can take time to maintain and make historical records harder to retrieve. This application stores student details and attendance in a central MySQL database. Authorized users can manage students, mark attendance as Present or Absent by subject and date, and view subject-wise summaries. Java Swing provides the graphical interface, JDBC connects the application to MySQL, and the report module calculates attendance percentages. This is an educational prototype that can later be extended with password hashing, roles, notifications and report exports.

2. Problem Statement

Manual attendance can require repeated paperwork, separate calculations and time-consuming searches. The project aims to make student records and attendance entries organized, searchable and easier to update.

3. Objectives

- Store student details in MySQL.
- Provide a login screen.
- Record Present/Absent status by student, subject and date.
- Prevent duplicate entries for the same student/date/subject.
- Calculate subject-wise attendance percentages.
- Reduce manual work and simplify access to records.

4. Scope

This prototype covers login, student management, attendance entry and attendance reports. Biometric devices, notifications, cloud hosting and production-grade account security are outside the current scope.

5. Technology Stack

Java 17+ provides application logic; Java Swing provides the desktop UI; JDBC connects Java to MySQL; MySQL stores user, student and attendance records; Maven manages the JDBC dependency and build.

6. Modules

Login: validates demo credentials. Student Management: add, view, update and delete student records. Attendance Management: records status for a selected student, subject and date. Reports: displays Present count, total records and percentage by student and subject. Database: stores records with primary keys, a foreign key and a uniqueness constraint.

7. Database Design

app_users: id, username, password and role. students: id, roll_no, full_name, department and year_level. attendance: id, student_id, attendance_date, subject and status. The attendance student_id references students.id. A unique constraint on student_id, attendance_date and subject prevents duplicate records.

8. Attendance Formula

Attendance Percentage = (Number of Present records / Total attendance records) × 100. Example: 8 Present records out of 10 total records gives 80%. The application calculates this separately for each subject.

9. Setup and Execution

1. Install JDK 17+, MySQL Server and Maven. 2. Start MySQL and execute database/schema.sql. 3. Edit Main.java and replace YOUR_MYSQL_PASSWORD; adjust DB_USER if needed. 4. Open a terminal in the folder containing pom.xml. 5. Run `mvn clean compile`. 6. Run `mvn exec:java`. 7. Sign in using admin / admin123.

10. Demonstration Procedure

Sign in. Add a student in the Students tab. Open Mark Attendance, choose a student, enter a subject and date as YYYY-MM-DD, choose Present or Absent, and save. Add another record so the report has data. Open Reports and choose Show All or search by roll number. Demonstrate updating a student record.

11. Testing Checklist

Login with valid credentials: dashboard opens. Invalid credentials: error message. Add a unique student: row appears. Duplicate roll number: database rejects it. Save attendance: record is stored. Save the same student/date/subject again: status updates. Open reports: counts and percentages display.

12. Limitations and Future Enhancements

This is a classroom prototype. The sample password is stored as plain text and must not be used in production. Future work could add password hashing, faculty/admin roles, date filters, CSV/PDF exports, notifications, charts, backups and QR-code or biometric attendance.

13. Conclusion

The project demonstrates how Java Swing and MySQL can be combined to manage student details, attendance entries and subject-wise summaries. It reduces reliance on paper registers and makes attendance information easier to retrieve. Its modular structure provides a foundation for further academic development.

14. References

Java: https://docs.oracle.com/en/java/
MySQL: https://dev.mysql.com/doc/
MySQL Connector/J: https://dev.mysql.com/doc/connector-j/en/
Apache Maven: https://maven.apache.org/